Linear First-Order Equations & the Integrating Factor

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Standard Linear Form and Integrating Factor**

Consider the second-order linear non-homogeneous differential equation: $\frac{d^2y}{dx^2} + 4y = \sec(2x)$. Suppose a particular solution is found using the method of variation of parameters. Let the fundamental set of solutions for the associated homogeneous equation be $y_1(x) = \cos(2x)$ and $y_2(x) = \sin(2x)$. If the particular solution is expressed in the form $y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x)$, determine the value of $u_2'(x)$.

The given differential equation is $\frac{d^2y}{dx^2} + 4y = \sec(2x)$. The associated homogeneous equation is $\frac{d^2y}{dx^2} + 4y = 0$. The characteristic equation is $r^2 + 4 = 0$, which has roots $r = \pm 2i$. Thus, the fundamental set of solutions for the homogeneous equation is $y_1(x) = \cos(2x)$ and $y_2(x) = \sin(2x)$. We use the method of variation of parameters to find a particular solution of the form $y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x)$. The Wronskian of y_1 and y_2 is.

2 **[Foundational]** • **Target: Solving Linear First-Order IVPs**

Consider the system of linear first-order differential equations given by: $(x'; y') = (1, 2; 3, 0)(x; y)$. Find the eigenvalues of the coefficient matrix.

The given system is of the form $\mathbf{x}' = A\mathbf{x}$, where $A = (1, 2; 3, 0)$ is the coefficient matrix. To find the eigenvalues, we solve the characteristic equation $\det(A - \lambda I) = 0$, where I is the identity matrix and λ represents the eigenvalues. $A - \lambda I = (1, 2; 3, 0) - \lambda(1, 0; 0, 1) = (1 - \lambda, 2; 3, -\lambda)$ det $(A - \lambda I) = (1 - \lambda)(-\lambda) - (2)(3) = -\lambda + \lambda^2 - 6 = \lambda^2 - \lambda - 6$ Sett.

3 **[Foundational]** • **Target: Mixing and Tank Applications**

Solve the first-order linear initial value problem $x\frac{dy}{dx} + 3y = \frac{\sin x}{x^2}$ for $x > 0$ with $y(\pi) = 0$. Find $\lim_{x \to 0^+} y(x)$.

Step 1: Standard form: Divide by x: $\frac{dy}{dx} + \frac{3}{x}y = \frac{\sin x}{x^3}$. Step 2: Integrating factor: $\mu(x) = \exp\left(\int \frac{3}{x}\,dx\right) = e^{3\ln x} = x^3$. Step 3: Multiply and integrate: $\frac{d}{dx}[x^3 y] = x^3\left(\frac{\sin x}{x^3}\right) = \sin x \Rightarrow x^3 y = -\cos x + C$. Step 4: Initial condition $y(\pi) = 0$: $\pi^3(0) = -\cos\pi + C \Rightarrow 0 = -(-1) + C \Rightarrow C = -1$. Thus: $y(x) = \frac{-\cos x - 1}{x^3} = -\frac{1 + \cos x}{x^3}$. Step 5: Limit as.

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4 [Foundational] • Target: Transient and Steady-State Solutions

Consider the second-order linear homogeneous differential equation $y'' - 2y' + y = 0$. \\ If one of the fundamental solutions is $y_1(x) = e^x$, and the Wronskian of two linearly independent solutions $y_1(x)$ and $y_2(x)$ is given by $W(y_1, y_2)(x) = 5e^{2x}$, find the unknown parameter k in $y_2(x) = kxe^x$.

The given differential equation is $y'' - 2y' + y = 0$. The characteristic equation is $r^2 - 2r + 1 = 0$, which factors as $(r-1)^2 = 0$. Thus, the roots are repeated: $r_1 = r_2 = 1$. \\ One fundamental solution is given as $y_1(x) = e^x$. For a second-order linear homogeneous equation $y'' + p(x)y' + q(x)y = 0$ with a known solution $y_1(x)$, the Wronskian is given by $W(y_1, y_2)(x) = y_1(x)y_2'(x) - y_1'(x)y_2(x) = Ce^{-\int p(x)dx}$. \\ In our equation, $p(x) = -2$. So, $-\int p(x)dx = -\int(-2)dx = 2x$..

5 [Foundational] • Target: Standard Linear Form and Integrating Factor

Consider the linear first-order differential equation $\frac{dy}{dx} + 2y = 6$. If the initial condition is $y(0) = 1$, find the value of $y(1)$.

The given differential equation is a linear first-order ODE of the form $\frac{dy}{dx} + P(x)y = Q(x)$, where $P(x) = 2$ and $Q(x) = 6$. The integrating factor $\mu(x)$ is given by $\mu(x) = e^{\int P(x)dx}$. Calculating the integrating factor: $\mu(x) = e^{\int 2dx} = e^{2x}$. Multiplying the differential equation by the integrating factor: $e^{2x}\frac{dy}{dx} + 2e^{2x}y = 6e^{2x}$. The left side is the derivative of the product $y\mu(x)$: $\frac{d}{dx}(ye^{2x}) = 6e^{2x}$. Integrating both sides with respect t.

6 [Foundational] • Target: Solving Linear First-Order IVPs

Consider the first-order linear differential equation $\frac{dy}{dx} + 2y = 4$. If the initial condition is $y(0) = 1$, find the value of $y(1)$.

The given differential equation is a first-order linear ODE of the form $\frac{dy}{dx} + P(x)y = Q(x)$. Here, $P(x) = 2$ and $Q(x) = 4$. The integrating factor is $I(x) = e^{\int P(x)dx} = e^{\int 2dx} = e^{2x}$. Multiplying the differential equation by the integrating factor, we get: $e^{2x}\frac{dy}{dx} + 2e^{2x}y = 4e^{2x}$. The left side is the derivative of $y \cdot I(x)$: $\frac{d}{dx}(ye^{2x}) = 4e^{2x}$. Integrating both sides with respect to x.

7 [Foundational] • Target: Mixing and Tank Applications

Consider the system of linear differential equations: $\frac{dx}{dt}, = ax + by; \frac{dy}{dt}, = cx + dy$ where a, b, c, d are constants. The characteristic equation for the system is $\lambda^2 - (a+d)\lambda + (ad - bc) = 0$. We are given that the eigenvalues of the system are $\lambda_1 = 2$ and $\lambda_2 = -1$. The trace of the coefficient matrix, $T = a + d$, is 1. Determine the value of the determinant of the coefficient matrix, $D = ad - bc$.

The characteristic equation of a 2x2 system of linear differential equations $\frac{dx}{dt}, = ax + by; \frac{dy}{dt}, = cx + dy$ is given by $\lambda^2 - (a+d)\lambda + (ad - bc) = 0$. This can be written in terms of the trace (T = a+d) and determinant (D = ad-bc) of the coefficient matrix as $\lambda^2 - T\lambda + D = 0$. We are given that the eigenvalues are $\lambda_1 = 2$ and $\lambda_2 = -1$. The sum of the roots of a quadratic equation $p\lambda^2 + q\lambda + r = 0$ is $-q/p$, and the product of the roots i.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Transient and Steady-State Solutions**

Consider the second-order nonhomogeneous linear differential equation $y'' - 2y' + y = e^x \sin(x)$. Let $y_p(x)$ be a particular solution obtained using the method of undetermined coefficients. Determine the value of $y_p''(\pi) - 2y_p'(\pi) + y_p(\pi)$.

The problem asks for the value of the left-hand side of the differential equation $y'' - 2y' + y = e^x \sin(x)$ when evaluated at a particular solution $y_p(x)$. By definition, a particular solution $y_p(x)$ must satisfy the nonhomogeneous differential equation for all x. Therefore, when we substitute $y_p(x)$ into the differential equation, the left-hand side must equal the right-hand side. Specifically, for any particular solution $y_p(x)$, it must be true that $y_p''(x) - 2y_p'(x) + y_p(x) = e^x \sin(x)$ for a.

9 **[Intermediate]** • **Target: Standard Linear Form and Integrating Factor**

Consider the nonhomogeneous second-order linear differential equation $y'' - 5y' + 6y = 12x + 8$. $y_h(x) = c_1 e^{2x} + c_2 e^{3x}$ is the general solution to the corresponding homogeneous equation. Find the particular solution $y_p(x)$ using the method of undetermined coefficients, and then determine the value of the constant A such that $y_p(x) = Ax + B$ for some constant B.

The given nonhomogeneous differential equation is $y'' - 5y' + 6y = 12x + 8$. The form of the nonhomogeneous term is $g(x) = 12x + 8$, which is a polynomial of degree 1. Thus, we assume a particular solution of the form $y_p(x) = Ax + B$. We need to find the first and second derivatives of $y_p(x)$: $y_p'(x) = A$ $y_p''(x) = 0$ Substitute these into the differential equation: $0 - 5(A) + 6(Ax + B) = 12x + 8$ $-5A + 6Ax + 6B = 12x + 8$ $6Ax + (6B - 5A) = 12x + 8$ Equating the coefficients of like powers of x: For.

10 **[Intermediate]** • **Target: Solving Linear First-Order IVPs**

Find all equilibrium solutions of $\frac{dy}{dt} = y^2 - 4$, classify each as an attractor (sink) or repeller (source), and determine $\lim_{t \to \infty} y(t)$ for the solution with $y(0) = 0$.

Step 1: Locate equilibria: Set $y^2 - 4 = 0 \Rightarrow y_1 = -2,\ y_2 = 2$. Step 2: Classify stability: Let $f(y) = y^2 - 4$. Then $f'(y) = 2y$. At $y = -2$: $f'(-2) = -4 < 0 \Rightarrow y = -2$ is a stable attractor (sink). At $y = 2$: $f'(2) = +4 > 0 \Rightarrow y = 2$ is an unstable repeller (source). Step 3: Trajectory with $y(0) = 0$: Since $y(0) = 0$ lies in the interval $(-2, 2)$, $f(y) = y^2 - 4 < 0$, so $y(t)$ is monotonically decreasing towards the lower stable equilibrium. Therefore.

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11 **[Intermediate]** • **Target: Mixing and Tank Applications**

Solve the initial value problem $\frac{dy}{dx} = 2xy^2$ with $y(0) = 1$. Find $y(1/2)$ and state its maximal interval of valid existence.

Step 1: Separate variables: $\frac{dy}{y^2} = 2x\,dx$. Step 2: Integrate: $-\frac{1}{y} = x^2 + C$. Step 3: Initial condition: $-\frac{1}{1} = 0 + C \Rightarrow C = -1$. Thus: $-\frac{1}{y} = x^2 - 1 \Rightarrow y(x) = \frac{1}{1-x^2}$. At $x = 1/2$: $y(1/2) = \frac{1}{1-1/4} = \frac{1}{3/4} = \frac{4}{3}$. Step 4: Maximal interval: Vertical asymptotes occur at $x = \pm 1$. The maximal open interval containing the initial point $x_0 = 0$ is $(-1, 1)$. As $x \to 1^-$, $y(x) \to +\infty$ (finite-t.

12 **[Intermediate]** • **Target: Transient and Steady-State Solutions**

Solve the second-order homogeneous initial value problem $y'' - 7y' + 10y = 0$ with $y(0) = 1$ and $y'(0) = -1$. Find $y(1)$.

Step 1: Auxiliary roots: $r^2 - 7r + 10 = (r-2)(r-5) = 0 \Rightarrow r_1 = 2, \quad r_2 = 5$. General solution: $y(x) = C_1 e^{2x} + C_2 e^{5x}$. Step 2: Derivative: $y'(x) = 2C_1 e^{2x} + 5C_2 e^{5x}$. Step 3: Apply initial conditions: $\{C_1 + C_2 = 1; 2C_1 + 5C_2 = -1$ From first: $C_1 = 1 - C_2$. Substitute: $2(1 - C_2) + 5C_2 = 2 + 3C_2 = -1 \Rightarrow 3C_2 = -3 \Rightarrow C_2 = -1$, $C_1 = 2$. Step 4: Particular solution: $y(x) = 2e^{2x} - e^{5x}, \qquad y(1) = 2e^2 - e^5$.

13 **[Intermediate]** • **Target: Standard Linear Form and Integrating Factor**

For the differential equation with a regular singular point at $x_0 = 0$: $2x^2y'' + xy' - (x+1)y = 0,$ find the indicial equation, determine the two roots $r_1 \geq r_2$, and find the first three non-zero terms of the Frobenius series solution corresponding to r_1.

Step 1: Indicial equation: $p_0 = \lim_{x\to 0} xP(x) = \lim_{x\to 0} x\left(\frac{x}{2x^2}\right) = \frac{1}{2}, \qquad q_0 = \lim_{x\to 0} x^2Q(x) = \lim_{x\to 0} x^2\left(-\frac{x+1}{2x^2}\right) = -\frac{1}{2}.$ $F(r) = r(r-1) + \frac{1}{2}r - \frac{1}{2} = r^2 - \frac{1}{2}r - \frac{1}{2} = 0 \Rightarrow 2r^2 - r - 1 = (2r+1)(r-1) = 0.$ Roots: $r_1 = 1$, $r_2 = -1/2$. Difference $r_1 - r_2 = 3/2 otin \mathbb{Z}$. Step 2: Frobenius series for $r_1 = 1$.

14 **[Intermediate]** • **Target: Solving Linear First-Order IVPs**

Solve the defective linear system $\mathbf{x}' = (1, -1; 1, 3)\mathbf{x}$ with $\mathbf{x}(0) = (2; 1)$.

Step 1: Eigenvalues: $\det(1-\lambda, -1; 1, 3-\lambda) = (1-\lambda)(3-\lambda) + 1 = \lambda^2 - 4\lambda + 4 = (\lambda - 2)^2 = 0 \Rightarrow \lambda = 2.$ Step 2: Eigenvector: $(A - 2I)\mathbf{v}_1 = (-1, -1; 1, 1)\mathbf{v}_1 = \mathbf{0} \Rightarrow \mathbf{v}_1 = (1; -1)$. Step 3: Generalized eigenvector: Solve $(A - 2I)\mathbf{v}_2 = \mathbf{v}_1$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 **[Challenging]** • **Target: Mixing and Tank Applications** • **Cross-Lesson: 5.01 Variation of Parameters**

Find the power series recurrence relation about $x_0 = 0$ for Hermite's differential equation with parameter 2: $y'' - 2xy' + 4y = 0, \quad y(0) = 1, \quad y'(0) = 0.$ Prove that the solution terminates to a quadratic polynomial and state the polynomial.

Step 1: Series substitution: $\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} - 2\sum_{n=1}^{\infty} n a_n x^n + 4\sum_{n=0}^{\infty} a_n x^n = 0.$ Shift first index: $\sum_{k=0}^{\infty} (k+2)(k+1)a_{k+2}x^k - \sum_{k=0}^{\infty} 2k a_k x^k + \sum_{k=0}^{\infty} 4a_k x^k = 0.$ Step 2: Recurrence relation: $(k+2)(k+1)a_{k+2} - 2(k-2)a_k = 0 \Rightarrow a_{k+2} = \frac{2(k-2)}{(k+2)(k+1)}a_k.$ Step 3: Initial conditions: $a_0 = 1$, $a_1 = 0$. Since $a_1 = 0$, all odd coefficients vanish: $a_3 = a_5 = a_7 = \ \ldots \ = 0$. For $k = 0$.

16 **[Challenging]** • **Target: Transient and Steady-State Solutions** • **Cross-Lesson: 1.01 Separable Equations**

Find the general solution of the linear system $\mathbf{x}' = (1, 3; 3, 1)\mathbf{x}$. Classify the equilibrium at the origin and determine the cartesian equation of the stable separatrix line.

Step 1: Characteristic polynomial: $\det(1-\lambda, 3; 3, 1-\lambda) = (1-\lambda)^2 - 9 = \lambda^2 - 2\lambda - 8 = (\lambda - 4)(\lambda + 2) = 0.$ $\lambda_1 = 4, \qquad \lambda_2 = -2.$ Since $\Delta = \det(A) = -8 < 0$, the origin is a saddle point. Step 2: Eigenvectors: For $\lambda_1 = 4$: $(-3, 3; 3, \ -3)\mathbf{v} = \mathbf{0} \Rightarrow \mathbf{v}_1 = (1; 1)$. For $\lambda_2 = -2$.

17 **[Challenging]** • **Target: Standard Linear Form and Integrating Factor** • **Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations**

Solve the nonlinear Bernoulli differential equation $\frac{dy}{dx} - \frac{1}{x}y = 2x^2 y^3$ for $x > 0$ with $y(1) = 1$. Find $y(2)$.

Step 1: Bernoulli transformation: Here $n = 3$. Divide by y^3: $y^{-3}y' - \frac{1}{x}y^{-2} = 2x^2$. Substitute $v = y^{1-n} = y^{-2} \Rightarrow v' = -2y^{-3}y' \Rightarrow y^{-3}y' = -\frac{1}{2}v'$. Substituting yields: $-\frac{1}{2}v' - \frac{1}{x}v = 2x^2 \Rightarrow v' + \frac{2}{x}v = -4x^2$. Step 2: Integrating factor for v: $\mu(x) = e^{\int 2/x\,dx} = x^2$. Multiply and integrate.

18 **[Challenging]** • **Target: Solving Linear First-Order IVPs** • **Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients**

Solve the Cauchy-Euler differential equation $x^2y'' - xy' + 5y = 0$ for $x > 0$ with $y(1) = 2$ and $y'(1) = 4$.

Step 1: Auxiliary equation: Substitute $y = x^m$: $m(m-1) - m + 5 = m^2 - 2m + 5 = 0 \Rightarrow m = \frac{2 \pm \sqrt{4-20}}{2} = 1 \pm 2i.$ For complex roots $\alpha \pm i\beta$, the real fundamental basis is $\{x^\alpha \cos(\beta \ln x), x^\alpha \sin(\beta \ln x)\}$. General solution: $y(x) = x(C_1 \cos(2\ln x) + C_2 \sin(2\ln x))$. Step 2: Apply $y(1) = 2$: Since $\ln 1 = 0$: $y(1) = 1(C_1 \cos 0 + 0) = C_1 = 2$. Step 3: Derivative.

19 **[Challenging]** • **Target: Mixing and Tank Applications** • **Cross-Lesson: 5.01 Variation of Parameters**

Solve the undamped resonant initial value problem $y'' + 9y = 6\cos(3t)$ with $y(0) = 0$ and $y'(0) = 1$. Identify the secular growth term.

Step 1: Homogeneous: $r^2 + 9 = 0 \Rightarrow y_c = C_1 \cos 3t + C_2 \sin 3t$. Step 2: Trial with resonance: Forcing $6\cos 3t$ matches y_c (multiplicity $s = 1$). Trial: $y_p(t) = t(A\cos 3t + B\sin 3t)$. Differentiating and substituting: $y_p'' + 9y_p = -6A\sin 3t + 6B\cos 3t = 6\cos 3t \Rightarrow B = 1, \quad A = 0.$ Thus $y_p(t) = t\sin(3t)$. Step 3: Apply initial conditions: $y(t) = C_1 \cos 3t + C_2 \sin 3t + t\sin 3t$. $y(0) = C_1 = 0$..

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20 [Challenging] • Target: Transient and Steady-State Solutions • Cross-Lesson: 1.01 Separable Equations

Find the general solution of $y'' - 4y' + 3y = 2x + e^{2x}$.

Step 1: Homogeneous: $r^2 - 4r + 3 = (r-1)(r-3) = 0 \Rightarrow y_c = C_1 e^x + C_2 e^{3x}$. Step 2: Particular trial form: Neither $2x$ nor e^{2x} is in y_c, so trial is $y_p = (Ax + B) + Ce^{2x}$. Step 3: Match polynomial coefficients: $y_{p1} = Ax + B \Rightarrow y'_{p1} = A, y''_{p1} = 0$. $0 - 4A + 3(Ax + B) = 3Ax + (3B - 4A) = 2x \Rightarrow 3A = 2 \Rightarrow A = 2/3$, $3B - 4(2/3) = 0 \Rightarrow 3B = 8/3 \Rightarrow B = 8/9$. Step 4: Match exponential.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Standard Linear Form and Integrating Factor

Write the differential equation $x\frac{dy}{dx} + 2y = 4x^2$ in standard form $\frac{dy}{dx} + P(x)y = Q(x)$ for $x > 0$.

a. $\frac{dy}{dx} + \frac{2}{x}y = 4x$ ✓

b. $\frac{dy}{dx} + 2y = 4x$

c. $\frac{dy}{dx} + \frac{2}{x}y = 4x^2$

d. $\frac{dy}{dx} = 4x - 2y$

Divide by the leading coefficient x: $\frac{dy}{dx} + \frac{2}{x}y = 4x$.

22 [Foundational] • Target: Standard Linear Form and Integrating Factor

Compute the integrating factor $I(x) = e^{\int P(x)dx}$ for $\frac{dy}{dx} + \frac{2}{x}y = 4x$.

a. x^2 ✓

b. $2\ln x$

c. e^{2x}

d. x

$I(x) = e^{\int (2/x)dx} = e^{2\ln x} = e^{\ln(x^2)} = x^2$.

23 [Foundational] • Target: Standard Linear Form and Integrating Factor

Multiply the standard ODE by $I(x) = x^2$ to solve $\frac{d}{dx}[x^2 y] = 4x^3$.

a. $y = x^2 + \frac{C}{x^2}$ ✓

b. $y = 4x^2 + C$

c. $y = x^4 + C$

d. $y = x^2 + C$

Integrate: $x^2 y = \int 4x^3 dx = x^4 + C \Rightarrow y = x^2 + \frac{C}{x^2}$.

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24 [Foundational] • Target: Solving Linear First-Order IVPs

Solve the IVP $\frac{dy}{dx} + y = e^{2x}$ with $y(0) = 1$.

a. $y = \frac{1}{3}e^{2x} + \frac{2}{3}e^{-x}$ ✓

b. $y = e^{2x}$

c. $y = \frac{1}{3}e^{2x} + e^{-x}$

d. $y = \frac{1}{2}e^{2x} + \frac{1}{2}e^{-x}$

$I(x) = e^x$. $\frac{d}{dx}[e^x y] = e^{3x} \Rightarrow e^x y = \frac{1}{3}e^{3x} + C \Rightarrow y = \frac{1}{3}e^{2x} + Ce^{-x}$. Using $y(0) = 1 \Rightarrow 1 = 1/3 + C \Rightarrow C = 2/3$. Thus $y = \frac{1}{3}e^{2x} + \frac{2}{3}e^{-x}$.

25 [Foundational] • Target: Solving Linear First-Order IVPs

Solve $\frac{dy}{dx} - 3y = 0$ with $y(0) = 4$.

a. $y = 4e^{3x}$ ✓

b. $y = 4e^{-3x}$

c. $y = 3e^{4x}$

d. $y = 4 + 3x$

This is a homogeneous linear equation with integrating factor e^{-3x}: $y = Ce^{3x}$. With $y(0) = 4$, $y = 4e^{3x}$.

26 [Foundational] • Target: Solving Linear First-Order IVPs

What is the domain of validity for the solution to $x\frac{dy}{dx} + y = 1$ with $y(1) = 2$?

a. $(0, \infty)$ ✓

b. $(-\infty, \infty)$

c. $[1, \infty)$

d. $(-\infty, 0)$

Standard form $\frac{dy}{dx} + \frac{1}{x}y = \frac{1}{x}$ has coefficients discontinuous at $x = 0$. By the Existence and Uniqueness Theorem, the solution through $x_0 = 1 > 0$ exists uniquely on $(0, \infty)$.

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27 [Foundational] • Target: Mixing and Tank Applications

A tank contains 100 L of water and 10 kg of dissolved salt. Pure water flows in at 3 L/min, and the stirred mixture drains at 3 L/min. What is the differential equation for the salt amount $A(t)$ in kg?

a. $\frac{dA}{dt} = -\frac{3}{100}A$ ✓

b. $\frac{dA}{dt} = 3 - \frac{3}{100}A$

c. $\frac{dA}{dt} = -3A$

d. $\frac{dA}{dt} = 10 - \frac{3}{100}A$

Rate in $= 0$ (pure water). Rate out $= 3 \times \frac{A}{100} = \frac{3}{100}A$. Differential equation: $\frac{dA}{dt} = -0.03A$.

28 [Foundational] • Target: Mixing and Tank Applications

For the tank above, what is the amount of salt remaining after t minutes?

a. $A(t) = 10e^{-0.03t}$ ✓

b. $A(t) = 10 - 0.03t$

c. $A(t) = 10e^{-3t}$

d. $A(t) = 10e^{0.03t}$

Separating variables: $A(t) = A(0)e^{-0.03t} = 10e^{-0.03t}$ kg.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Mixing and Tank Applications

If salt solution enters at 2 kg/L at a rate of 3 L/min, what is the inflow rate of salt in kg/min?

a. 6 kg/min ✓

b. 2 kg/min

c. 3 kg/min

d. 1.5 kg/min

Rate in = concentration × flow rate = (2 kg/L) × (3 L/min) = 6 kg/min.

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30 [Intermediate] • Target: Transient and Steady-State Solutions

In the solution $y(t) = 5 + 4e^{-2t}$, which part represents the transient response and which represents the steady-state?

a. 5 **is steady-state,** $4e^{-2t}$ **is transient** ✓

b. $4e^{-2t}$ is steady-state, 5 is transient

c. Both are steady-state

d. The entire response is transient

Because $\lim_{t\to\infty} 4e^{-2t} = 0$, that term dies out and is called transient; the permanent value 5 is the steady-state response.

31 [Intermediate] • Target: Transient and Steady-State Solutions

What is the time constant τ for the decaying exponential $e^{-t/\tau} = e^{-5t}$?

a. $\tau = 0.2$ seconds ✓

b. $\tau = 5$ seconds

c. $\tau = 1$ second

d. $\tau = 2.5$ seconds

Matching $e^{-t/\tau}$ with e^{-5t} gives $\frac{1}{\tau} = 5 \Rightarrow \tau = \frac{1}{5} = 0.2$.

32 [Intermediate] • Target: Transient and Steady-State Solutions

After how many time constants ($t = n\tau$) has a decaying transient decayed to less than 1% of its initial value?

a. 5 **time constants (**$e^{-5} \approx 0.0067$**)** ✓

b. 1 time constant

c. 3 time constants

d. 10 time constants

At $t = 5\tau$, $e^{-5} \approx 0.0067 < 0.01$ (less than 1% remaining), which is the standard engineering threshold for steady-state.

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33 [Intermediate] • Target: Standard Linear Form and Integrating Factor

Consider the first-order linear differential equation $\frac{dy}{dx} + P(x)y = Q(x)$. If the integrating factor for this equation is given by e^{x^2} and $Q(x) = 2xe^{x^2}$, find the value of the constant C in the general solution if $y(0) = 3$. The general solution is of the form $y = e^{-x^2}\left(\int Q(x)e^{-x^2}dx + C\right)$.

a. 1

b. **3** ✓

c. 4

d. 6

The given differential equation is in the form $\frac{dy}{dx} + P(x)y = Q(x)$. The integrating factor (IF) is given by $e^{\int P(x)dx}$. We are told that the integrating factor is e^{x^2}. Therefore, $\int P(x)dx = x^2$, which implies $P(x) = 2x$. The general solution to a first-order linear differential equation is given by $y \cdot \text{IF} = \int Q(x) \cdot \text{IF}\, dx + C$. Substituting the IF and $Q(x)$, we get: $ye^{x^2} = \int (2xe^{x^2})e^{-x^2}\, dx + C$ $ye^{x^2} = \int 2x\, dx + C$.

34 [Intermediate] • Target: Solving Linear First-Order IVPs

An RC circuit with $R = 4\,\Omega$ and $C = 0.5$ F is connected to $E(t) = 20\cos t$ V. What is the amplitude of the steady-state charge $q_{ss}(t)$?

a. $4\sqrt{5}$ C

b. 5 C

c. $\sqrt{10}$ C

d. $2\sqrt{5}$ C ✓

Circuit ODE: $4q' + 2q = 20\cos t \Rightarrow q' + 0.5q = 5\cos t$. Let $q_{ss} = A\cos t + B\sin t$. Substituting gives $(-A + 0.5B)\sin t + (B + 0.5A)\cos t = 5\cos t$. Solving $-A + 0.5B = 0$ and $B + 0.5A = 5$ yields $A = 2$ and $B = 4$. The amplitude of the steady-state charge is $\sqrt{A^2 + B^2} = \sqrt{4 + 16} = \sqrt{20} = 2\sqrt{5}$ C.

35 [Intermediate] • Target: Mixing and Tank Applications

An RL circuit has inductance $L = 5$ H and resistance $R = 15\,\Omega$. At $t = 0$, $i(0) = 0$ A. A constant voltage $E = 60$ V is applied. At what time t does the current reach 90% of its steady-state value?

a. $\frac{1}{5}\ln 10$

b. $3\ln 10$

c. $\frac{1}{3}\ln(10/9)$

d. $\frac{1}{3}\ln 10$ ✓

Circuit equation: $5i' + 15i = 60 \Rightarrow i' + 3i = 12$. Integrating factor: $\mu = e^{3t}$. Solving yields $i(t) = 4(1 - e^{-3t})$. The steady-state current is $i_{ss} = 4$ A. Reaching 90% means

$i(t) = 3.6 \text{ A} \Rightarrow 4(1 - e^{-3t}) = 3.6 \Rightarrow e^{-3t} = 0.1 \Rightarrow 3t = \ln 10 \Rightarrow t = \frac{1}{3}\ln 10$.

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36 [Intermediate] • Target: Transient and Steady-State Solutions

An RL circuit with $L = 1$ H and $R = 2\,\Omega$ has initial current $i(0) = 0$ A. The applied voltage is $E(t) = 10\sin(2t)$ V. What is the amplitude of the steady-state alternating current $i_{ss}(t)$?

a. $5\sqrt{2}$ A

b. $\frac{5}{2}$ A

c. $\frac{5\sqrt{2}}{2}$ A ✓

d. 5 A

Circuit equation: $i' + 2i = 10\sin(2t)$. Integrating factor: $\mu = e^{2t}$. Multiplying gives $(ie^{2t})' = 10e^{2t}\sin(2t)$. Integrating yields $ie^{2t} = \frac{5}{2}e^{2t}(\sin 2t - \cos 2t) + C \Rightarrow i_{ss}(t) = \frac{5}{2}\sin(2t) - \frac{5}{2}\cos(2t)$. The amplitude is $\sqrt{(5/2)^2 + (-5/2)^2} = \frac{5\sqrt{2}}{2}$ A.

37 [Intermediate] • Target: Standard Linear Form and Integrating Factor

An RL circuit has $L = 3$ H, $R = 9\,\Omega$, and electromotive force $E(t) = 18\cos(3t)$ V. What is the steady-state current $i_{ss}(t)$?

a. $\cos(3t) - \sin(3t)$

b. $\cos(3t) + \sin(3t)$ ✓

c. $2\cos(3t) + 2\sin(3t)$

d. $\frac{1}{2}(\cos 3t + \sin 3t)$

$3i' + 9i = 18\cos(3t) \Rightarrow i' + 3i = 6\cos(3t)$. $\mu = e^{3t}$. Multiplying gives $(ie^{3t})' = 6e^{3t}\cos(3t)$. Integrating yields $ie^{3t} = e^{3t}(\cos 3t + \sin 3t) + C \Rightarrow i(t) = \cos(3t) + \sin(3t) + Ce^{-3t}$. The steady-state component is $i_{ss}(t) = \cos(3t) + \sin(3t)$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Solving Linear First-Order IVPs • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

A mixing tank contains 300 liters of brine with 15 kg of salt. Brine containing 0.1 kg/L of salt flows in at 6 L/min, and the mixture drains out at 6 L/min. What is the salt concentration in the tank as $t \to \infty$?

a. 0.05 kg/L

b. 0.1 kg/L ✓

c. 0.2 kg/L

d. 0.15 kg/L

The salt mass obeys $S' + \frac{6}{300}S = 6(0.1) \Rightarrow S' + 0.02S = 0.6$. The steady-state amount of salt is $S_\infty = \frac{0.6}{0.02} = 30$ kg. Since the volume remains 300 L, the limiting concentration is $\frac{30}{300} = 0.1$ kg/L.

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39 [Challenging] • Target: Mixing and Tank Applications • Cross-Lesson: 5.01 Variation of Parameters

Consider the second-order linear homogeneous differential equation $y'' - 2y' + 5y = 0$. Let $y_1(x)$ and $y_2(x)$ be two linearly independent solutions. Suppose there exists a solution of the form $y(x) = c_1y_1(x) + c_2y_2(x)$ such that $y(0) = 1$ and $y'(0) = 2$. Furthermore, let the Wronskian of y_1 and y_2 be given by $W(y_1, y_2)(x) = e^{2x}$. Determine the value of the expression $y''(0) - 2y'(0) + 5y(0)$.

a. **0** ✓

b. e^2

c. 5

d. $2e^2$

The given differential equation is $y'' - 2y' + 5y = 0$. This is a second-order linear homogeneous differential equation. By definition, any solution $y(x)$ to this differential equation must satisfy the equation itself. We are given that $y(x)$ is a solution such that $y(0) = 1$ and $y'(0) = 2$. We are asked to find the value of $y''(0) - 2y'(0) + 5y(0)$. Since $y(x)$ is a solution to the differential equation $y'' - 2y' + 5y = 0$, by substituting $x = 0$ into the differential equation, we get.

40 [Challenging] • Target: Transient and Steady-State Solutions • Cross-Lesson: 1.01 Separable Equations

A 120-gallon tank initially holds 60 gallons of brine containing 30 lb of dissolved salt. Brine containing 1 lb/gal enters at 3 gal/min, while the well-stirred mixture leaves at 1 gal/min. How much salt is in the tank at the exact instant the tank becomes full?

a. 90 lb

b. $120 - 15\sqrt{2}$ lb ✓

c. $120 - 30\sqrt{2}$ lb

d. $120 - 15\sqrt{3}$ lb

Volume: $V(t) = 60 + 2t = 120 \Rightarrow t = 30$ min. $S' = 3(1) - 1\frac{S}{60 + 2t} \Rightarrow S' + \frac{1}{2(t + 30)}S = 3$. Integrating factor: $\mu = \sqrt{t + 30}$. Multiplying gives $(\sqrt{t + 30}S)' = 3\sqrt{t + 30}$. Integrating yields $\sqrt{t + 30}S = 2(t + 30)^{3/2} + C$. Using $S(0) = 30$ yields $\sqrt{30}(30) = 2(30\sqrt{30}) + C \Rightarrow C = -30\sqrt{30}$. At $t = 30$, $t + 30 = 60$, so.

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41 [Challenging] • Target: Standard Linear Form and Integrating Factor • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

A 500-gallon tank initially contains 200 gallons of brine with 40 lb of salt. Brine containing 0.5 lb/gal of salt enters at 6 gal/min, and the well-stirred mixture drains at 4 gal/min. What is the salt concentration in lb/gal when the tank contains 300 gallons?

a. $\frac{37}{45}$ lb/gal

b. $\frac{41}{90}$ lb/gal

c. $\frac{1}{2}$ lb/gal

d. $\frac{37}{90}$ lb/gal ✓

$V(t) = 200 + 2t = 300 \Rightarrow t = 50$ min. Salt equation: $S' + \frac{2}{t+100}S = 3$. Integrating factor: $\mu = (t+100)^2$. Multiplying gives $((t+100)^2 S)' = 3(t+100)^2 \Rightarrow (t+100)^2 S = (t+100)^3 + C$. Using $S(0) = 40$ yields $100^2(40) = 100^3 + C \Rightarrow C = -600{,}000$. At $t = 50$, $t + 100 = 150$, so $S(50) = 150 - \frac{600000}{22500} = 150 - \frac{80}{3} = \frac{370}{3}$ lb. The concentration is $c = \frac{370/3}{300} = \frac{37}{90}$ lb/gal.

42 [Challenging] • Target: Solving Linear First-Order IVPs • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

A tank contains 100 L of water in which 20 kg of salt is dissolved. Brine containing 0.3 kg/L of salt enters at 5 L/min, and the mixture leaves at 3 L/min. What is the mass of salt in the tank after 50 minutes?

a. $60 - \frac{5\sqrt{2}}{2}$ kg ✓

b. $60 - 5\sqrt{2}$ kg

c. $60 - \frac{5}{2}$ kg

d. $30 - \frac{5\sqrt{2}}{2}$ kg

$V(t) = 100 + 2t$. Salt equation: $S' + \frac{1.5}{t+50}S = 1.5$. Integrating factor: $\mu = (t+50)^{3/2}$. Multiplying gives $((t+50)^{3/2}S)' = 1.5(t+50)^{3/2}$. Integrating yields $(t+50)^{3/2}S = 0.6(t+50)^{5/2} + C$. Using $S(0) = 20$ yields $20(50)^{3/2} = 0.6(50)^{5/2} + C = 30(50)^{3/2} + C \Rightarrow C = -10(50)^{3/2}$. At $t = 50$, $t + 50 = 100$, so $S(50) = 0.6(100) - 10(50/100)^{3/2} = 60 - \frac{10}{2\sqrt{2}} = 60 - \frac{5\sqrt{2}}{2}$ kg.

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43 [Challenging] • Target: Mixing and Tank Applications • Cross-Lesson: 5.01 Variation of Parameters

A tank initially holds 100 gal of brine containing 20 lb of salt. Brine containing 1 lb/gal enters at 4 gal/min, and the well-stirred mixture drains out at 6 gal/min. How much salt is in the tank after 25 minutes?

a. 45 lb

b. 40 lb ✓

c. 35 lb

d. 50 lb

$V(t) = 100 - 2t$. Salt equation: $S' = 4(1) - 6\frac{S}{100-2t} \Rightarrow S' + \frac{3}{50-t}S = 4$. Integrating factor: $\mu = (50-t)^{-3}$. Multiplying gives $((50-t)^{-3}S)' = 4(50-t)^{-3}$. Integrating yields $(50-t)^{-3}S = 2(50-t)^{-2} + C \Rightarrow S(t) = 2(50-t) + C(50-t)^3$. Using $S(0) = 20$ gives $20 = 100 + C(50^3) \Rightarrow C(50^3) = -80$. At $t = 25$, $50 - t = 25 = 50/2$, so $S(25) = 2(25) + C(50^3)/8 = 50 - 80/8 = 40$ lb.

44 [Challenging] • Target: Transient and Steady-State Solutions • Cross-Lesson: 1.01 Separable Equations

A tank initially contains 100 L of pure water. Brine containing 2 kg/L of salt flows in at 4 L/min, and the thoroughly mixed liquid flows out at 2 L/min. What is the total mass of salt in the tank after 50 minutes?

a. 300 kg ✓

b. 250 kg

c. 350 kg

d. 200 kg

Changing volume: $V(t) = 100 + (4-2)t = 100 + 2t$. The salt mass obeys $S' = 4(2) - 2\frac{S}{100+2t} \Rightarrow S' + \frac{1}{t+50}S = 8$. Integrating factor: $\mu = t + 50$. Multiplying gives $((t+50)S)' = 8(t+50)$. Integrating gives $(t+50)S = 4(t+50)^2 + C$. Using $S(0) = 0$ yields $50(0) = 4(2500) + C \Rightarrow C = -10000$. Thus $S(t) = 4(t+50) - \frac{10000}{t+50}$. At $t = 50$, $S(50) = 4(100) - \frac{10000}{100} = 400 - 100 = 300$ kg.

45 [Challenging] • Target: Standard Linear Form and Integrating Factor • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

A tank initially contains 80 L of brine containing 16 kg of salt. Brine containing 0.5 kg/L enters at 6 L/min, and the mixture flows out at 8 L/min. What is the amount of salt in the tank after 20 minutes?

a. $\frac{37}{2}$ kg ✓

b. 19 kg

c. 18 kg

d. $\frac{35}{2}$ kg

$V(t) = 80 - 2t$. Salt equation: $S' + \frac{4}{40-t}S = 3$. Integrating factor: $\mu = (40-t)^{-4}$. Multiplying gives $((40-t)^{-4}S)' = 3(40-t)^{-4}$. Integrating gives $(40-t)^{-4}S = (40-t)^{-3} + C \Rightarrow S(t) = (40-t) + C(40-t)^4$. Using $S(0) = 16$ gives $16 = 40 + C(40^4) \Rightarrow C(40^4) = -24$. At $t = 20$, $40 - t = 20 = 40/2$, so $S(20) = 20 + C(40^4)/16 = 20 - 24/16 = 20 - 1.5 = 18.5 = \frac{37}{2}$ kg.

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